$$\frac{(\alpha+\beta)|00\rangle + (\alpha-\beta)|11\rangle}{\sqrt{2}}$$

before measurement

`lift`, then V

record 1 (safe, $g = |0\rangle$)

$$\frac{\alpha+\beta}{\sqrt{2}}\,|0\rangle \otimes |0\rangle$$

weight $|\alpha+\beta|^2/2$

record 0 (unsafe)

$$\frac{\alpha-\beta}{2}\,(|01\rangle + |10\rangle)$$

weight $|\alpha-\beta|^2/2$